



MDMC

Master in Data Management
and Curation

AREA
SCIENCE PARK
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Scientific Programming Environment Lecture I: UNIX

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Special thanks to Gianfranco Gallizia



Image credit: Thomas Schanz @ wikimedia.org



Goals

Objectives of the lecture:

- * Learn the history behind **UNIX** and **GNU/Linux**
- * Understand the fundamental **philosophy** of UNIX
- * Look at the **structure** of a UNIX-based machine



Prologue: why?

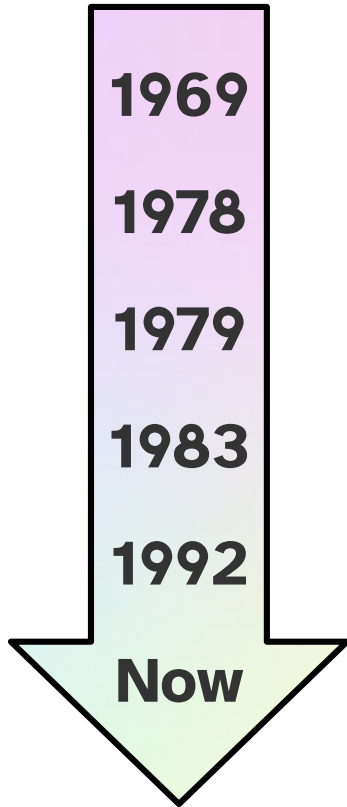
Why bother with UNIX and the terminal?

- * “The world runs on Linux”
- * Standard interface for **scientific computing**
- * Much handy when **automating** tasks

```
jacosposalvalaggio@area-pf2hdes3: ~/fonts — top
~/fonts
top - 16:51:15 up 6:32, 1 user, load average: 0.64, 0.76, 0.75
Tasks: 370 total, 1 running, 368 sleeping, 0 stopped, 1 zombie
%Cpu(s): 3.4 us, 1.2 sy, 0.0 ni, 95.2 id, 0.2 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15101.9 total, 306.5 free, 10085.7 used, 5816.9 buff/cache
MiB Swap: 4096.0 total, 4094.7 free, 1.3 used. 5016.2 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
 5010 jacospos+  20   0 7389404 364676 193076 S 20.9   2.4   65:35.67 gnome-shell
23389 jacospos+  20   0 2322692 179892 119456 S   2.6   1.2   1:47.73 ptysis
109836 jacospos+  20   0 2741248 212876 117224 S   2.0   1.4   0:09.39 Isolated Web Co
13824 jacospos+  20   0  13.1g  1.0g 420288 S   1.3   6.7 104:11.44 firefox
 9010 root       20   0 3904232 868464 212664 S   1.0   5.6   3:50.51 python3
20322 jacospos+  20   0 2632096 121356  95884 S   1.0   0.8   0:38.63 file:// Content
97329 jacospos+  20   0 2755284 684276 180924 S   1.0   4.4   4:13.21 soffice.bin
119696 jacospos+  20   0  13392   6220  4072 R   1.0   0.0   0:00.07 top
 2328 polkitd    20   0 389432  12812   8476 S   0.7   0.1   0:03.37 polkitd
46785 jacospos+  20   0 3118848 443368 124288 S   0.7   2.9   1:28.84 Isolated Web Co
116528 jacospos+  20   0 2621368  97112  73944 S   0.7   0.6   0:00.20 Isolated Servic
   80 root      -51   0     0     0     0 S   0.3   0.0   0:49.21 irq/9-acpi
1533 root      -51   0     0     0     0 S   0.3   0.0   0:00.63 irq/176-iwlwifi+
2311 message+  20   0  12184   9264  4592 S   0.3   0.1   0:17.97 dbus-daemon
4996 jacospos+  20   0 615880  11448  9944 S   0.3   0.1   0:00.65 gnome-session-s
7141 root      20   0 2266156 50204  32380 S   0.3   0.3   0:50.49 containerd
42893 jacospos+  20   0 3304792 483100 152508 S   0.3   3.1   5:02.00 Isolated Web Co
52675 root      -20   0     0     0     0 I   0.3   0.0   0:17.65 kworker/u33:1-+
53747 jacospos+  20   0 3451020 639828 127236 S   0.3   4.1   3:27.12 Isolated Web Co
57978 jacospos+  20   0 2718892 171376 107788 S   0.3   1.1   0:33.64 Isolated Web Co
```

A timeline of UNIX



1969

Development started at by Thompson, Ritchie et al.

1978

First release of BSD

1979

Commercial release of System V

1983

The GNU project is born

1992

The Linux kernel is release under GNU GPL

Now

Most servers and smartphones run (GNU/)Linux

Some faces



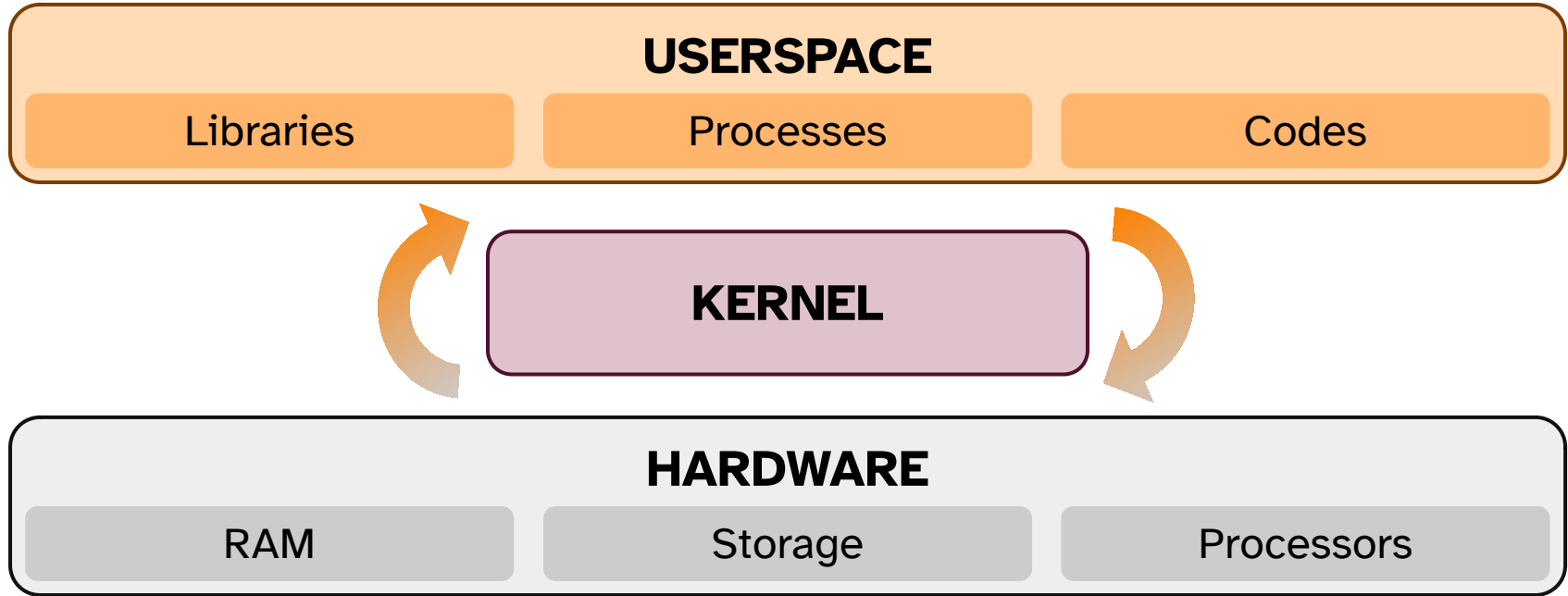
Image credits from left to right:
wikimedia.org, conati.gob.ve,
newyorker.com

The structure of UNIX

A UNIX machine operates by separating the **userspace** from the **kernel**:

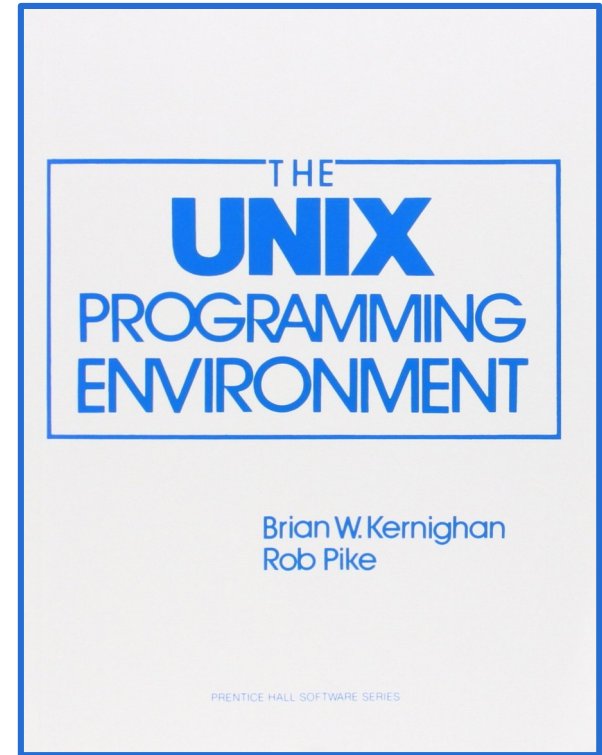
- * The **userspace** or userland is where the **code executed by the user** lives
- * The **kernel** is the **interface with hardware** that allocates the memory to be used by the processes in userland
- * No user code can allocate memory without asking the kernel

Userspace and kernel



The philosophy of UNIX

- * **Small programs** that work **together** are better than a big piece of code doing everything
- * **Everything is a file**
- * Files are organized in a **hierarchy of directories**
- * The **shell** is both a command and the command line interpreter



Working together

Big programs that do everything can become unwieldy, so it is **better to create small programs that do one thing and do it well.**

Then, you can **concatenate** them to complete complex tasks.

Example: count how many lines contain the word “UNIX” in two files

Concatenate the files → Select lines where “UNIX” appears → Count the number of lines

Working together – a quote

Even though the UNIX system introduces a number of innovative programs and techniques, no single program or idea makes it work well. Instead, what makes it effective is an approach to programming, a philosophy of using the computer. Although that philosophy can't be written down in a single sentence, at its heart is the idea that the power of a system comes more from the relationships among programs than from the programs themselves. Many UNIX programs do quite trivial tasks in isolation, but, combined with other programs, become general and useful tools.

- Kerninghan & Pike, *The UNIX Programming Environment*

Everything is a file

Everything in Linux is treated as a file. Types of files can be:

- * Ordinary files
- * Directories
- * Links
- * Devices
- * ...



Fig. 1: A file



Fig. 2: Also a file

Image credits: freeiconspng.com/img/1215, static.vecteezy.com

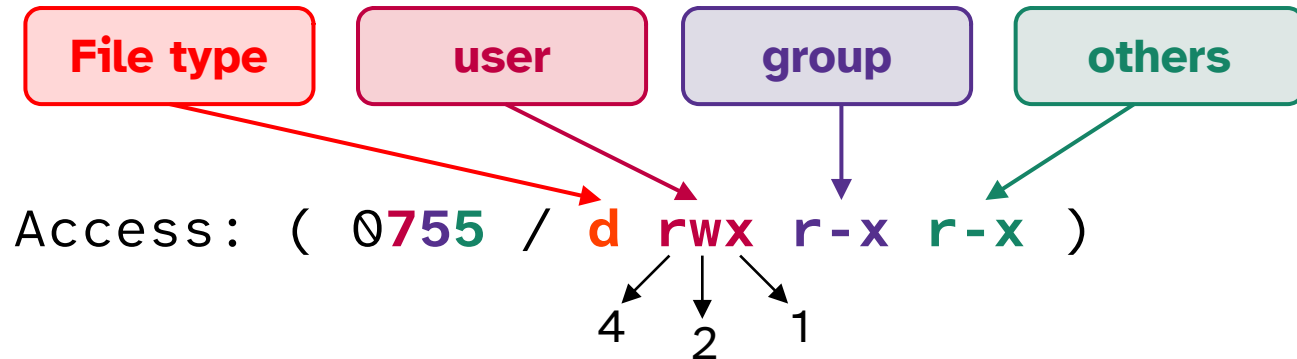
File metadata

Each file contains info on its **type**, **owner**, **creation date** and much more.

```
$ stat Desktop/  
  File: Desktop/  
  size: 4096          Blocks: 8          IO Block: 4096    directory  
Device: 252,1  Inode: 29231255    Links: 2  
Access: (0755/drwxr-xr-x)  Uid: ( 1001/jacoposalvalaggio)  Gid:  
( 1001/jacoposalvalaggio)  
Access: 2026-09-21 10:20:01.923384534 +0200  
Modify: 2026-05-27 15:25:20.644418049 +0200  
Change: 2026-05-27 15:25:20.644418049 +0200  
Birth: 2026-05-27 15:25:20.644418049 +0200
```

File ownership

- * Each file is owned by a **user** and a **group**
- * **Permissions** determine how the file can be accessed and can be **read** (r), **write** (w) and **execute** (x)



File permissions

Here are some example permissions:

- * Private file where user has all permissions:
700 / rwx - - - - -
- * Non-executable file shared with group members:
640 / rw- r - - - -
- * Free-for-all
777 / rwx rwx rwx

The FHS

All files in a UNIX system are organized in a hierarchy of folders:

- * “/”: the **root folder**, contains everything else
- * All directories (and files) can be identified by their **path**, that is the directories between them and root
- * Special directory names: `~/` , `./` , `../`

The hierarchy

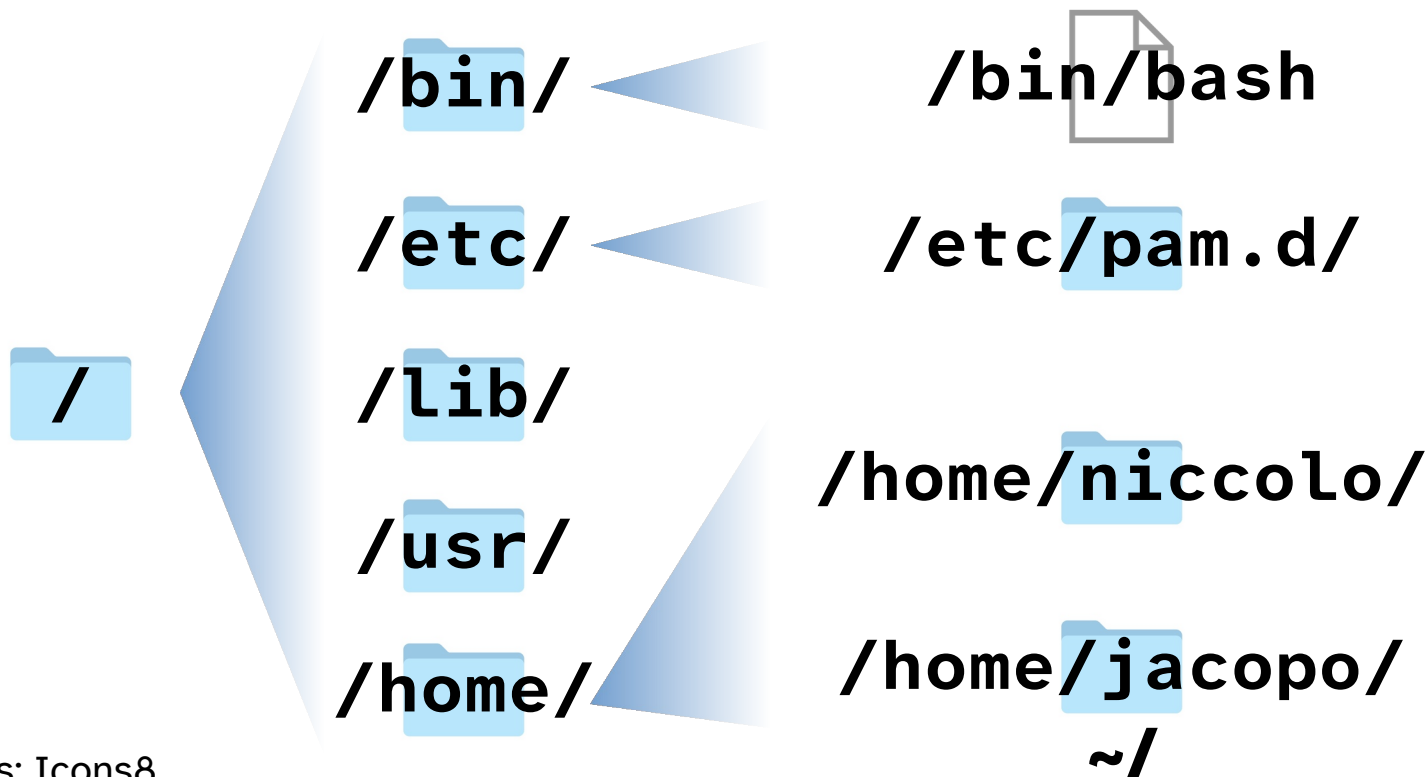


Image credits: Icons8

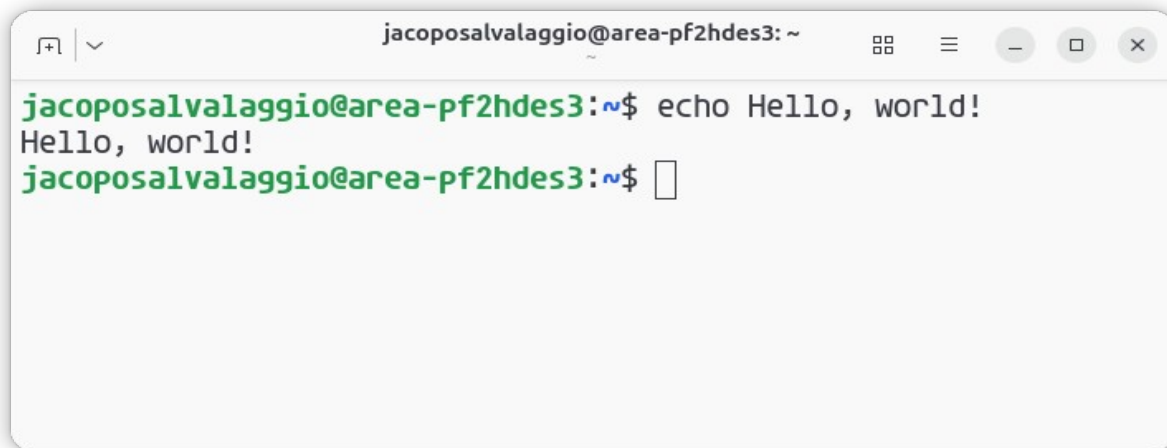
Relative paths



Image credits: Icons8

Shell

The **shell** is the interface with the UNIX system. Fundamentally, it is **both a command and the command line interpreter**.

A terminal window with a title bar showing the user 'jacoposalvalaggio@area-pf2hdes3' and the home directory '~'. The terminal displays a command 'echo Hello, world!' and its output 'Hello, world!'. The prompt is 'jacoposalvalaggio@area-pf2hdes3:~\$' and there is a cursor after the second prompt.

```
jacoposalvalaggio@area-pf2hdes3: ~  
jacoposalvalaggio@area-pf2hdes3:~$ echo Hello, world!  
Hello, world!  
jacoposalvalaggio@area-pf2hdes3:~$
```

Commands

Commands are passed to the shell to **perform actions** in the computer, e.g. enter a directory, list files, copy a file, etc.

The **shell itself is a command** in the sense that you can group a list of commands in a file (called **script**) and run it from the shell.

This is **not trivial**! In ancient systems commands in scripts were not necessarily the same as the ones in the console...

Our system: ORFEO

- * HPC system in **container**
- * Located in **Basovizza**
- * Used both by **researchers, students** and **companies**
- * Lots of varied pieces of hardware!



Image credits: AREA Science Park

Hardware

- * 40+ CPU nodes with **~3000 cores**
- * 7 GPU nodes with **~40 GPUs**
- * 1 GPU node for **LLM inference**
- * **5PB** of storage



Image credits: Massimo Luin



Image credit: Abel+Imray

Thank you for your attention :)

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